

# Changyu (Thomason) Zhao

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## EDUCATION

### Stanford University

*M.S. in Computer Science (Computer and Network Security)*

Sep. 2025 – Jun. 2027 (expected)

*Stanford, California*

### University of Wisconsin-Madison

*B.S. in Computer Science (Honors) and Mathematics*

Sep. 2021 – May 2025

*Madison, Wisconsin*

GPA: **3.95/4.0**, David Dewitt Undergraduate Scholarship, Dean's List (all semesters)

## EXPERIENCES

### UW-Madison Cybersecurity Operation Center

Feb. 2023 – May 2025

*Cybersecurity Student Intern Analyst*

*Madison, Wisconsin*

- Investigated **1000+** security incidents (malware, phishing, intrusions) using **Elastic SIEM**. Developed **threat hunting queries and detection rules** that reduced MTTR by **40%** and repeat infections by **25%**.
- Conducted monthly **IAM attestations** for **500+** accounts using automated scripts, remediating 3% toxic access in 24h.

### UW-Madison MadS&P Lab

Sep. 2024 – Aug. 2025

*Research Assistant, Advisor: Prof. Patrick McDaniel*

*Madison, Wisconsin*

- Conducted **comprehensive evaluation** of LibIHT against Intel Pin and DynamoRIO on real-world workloads, demonstrating **150x speedup**, **99%+ code coverage**, and **complete resistance** to anti-debugging techniques employed by malware.
- Established **reproducible benchmarking infrastructure** with 21 coreutils binaries and 10 malware specimens, automating performance metrics collection and visualization in **Python**. Paper accepted by **ACM CCS 2025 SURE** workshop.

### ASU SEFCOM Lab

May. 2024 – Sep. 2024

*Research Assistant, Advisor: Prof. Yan Shoshitaishvili and Ruoyu Wang*

*Tempe, Arizona*

- Co-designed Oxidizer, the first **Rust-focused decompiler** built atop angr, implementing control-flow recovery, function prototype inference, and macro inference to address the unique challenges of analyzing memory-safe compiled binaries.
- Developed **static type recovery engine** to extract Rust-specific type metadata from compiled binaries and integrates findings into decompilation, improving accuracy of data structure and function signature reconstruction for security analysis.

### Tencent Xuanwu Lab

Jul. 2023 – Aug. 2023

*Security Researcher Intern*

*Beijing, China*

- Architected **LibIHT**, a cross-platform dynamic binary tracing framework in **C/C++/Python** that leverages Intel LBR/BTS hardware features for control-flow tracing, achieving **150x speedup** over Intel Pin on large-scale program analysis.
- Designed unified **kernel driver architecture** for Linux and Windows with abstraction layers for OS and hardware, reducing code duplication by **40%** while standardizing trace session management and improving cross-platform maintainability.
- Implemented **user-space API library** and debugger plugin system for **GDB/WinDbg** integration, enabling security analysts to visualize execution traces, identify anomalous control-flow patterns, and accelerate vulnerability discovery workflows.

### ASU SEFCOM Lab

May. 2023 – Jul. 2023

*Research Assistant, Advisor: Prof. Yan Shoshitaishvili*

*Tempe, Arizona*

- Developed **LibHopper**, an automated exploitation primitive discovery tool combining debugger instrumentation with **symbolic execution** to identify exploitable memory corruption opportunities in library state structures.
- Constructed **exploitation primitives dataset** spanning critical C libraries (OpenSSL, OpenSSH, libcurl, libpng), providing structured risk assessments to inform security research in automated exploit generation and vulnerability scanning tooling.

## PROJECTS

### Router automated vulnerability discovery | IoT, Reverse Engineering, Web Security

May 2022 – Aug. 2022

- Developed static **automated vulnerability discovery** framework for IoT firmware with IDA plugin script. Uncovered **7 unfixed CVEs** in Tenda AC 18 router, including stack overflow and command injection vulnerabilities.
- Performed **dynamic analysis** using **QEMU** rehosting and **GDB** to craft PoC exploits for identified vulnerabilities.

## PRESENTATIONS & PUBLICATIONS

Yibo Liu, ..., **Changyu Zhao**, ... "Toward Concise and High-fidelity Rust Decompilation." (In submission).

**Changyu Zhao**, ... "LibIHT: A Hardware-Based Approach to Efficient and Evasion-Resilient Dynamic Binary Analysis". In *Proceedings of the 1st Workshop on Software Understanding and Reverse Engineering (SURE 2025)*. ACM, October 2025.

Black Hat Arsenal, **LIBIHT: A Cross-Platform Library for Accessing Intel Hardware Trace Features**, Las Vegas, NV, Aug. 2024

## EXTRACURRICULAR ACTIVITIES

### r3kapig CTF Team

Sep. 2021 – Present

*Team Member*

*International*

- Entered **DEFCON CTF world final** from 2022 to 2025, ranked 12th, 7th, 4th, and 5th respectively.
- Competed in **120+** CTF competitions, expert in kernel and user-space exploitation and reverse engineering.

## SKILLS

**Programming Languages:** C/C++, Python, Java, Rust, Go, Assembly Language (x86/x64, ARM, MIPS)

**Binary Security Tools:** IDA Pro/Ghidra, GDB/Windbg, Angr, Qemu, AFL++, Frida, Joern

**Web Security Tools:** Burpsuit, Wireshark, Metasploit Framework, Kali Linux